



Patient Profile

Saubhik Bhaumik, 27-year-old male

Summary of Results

Liver function tests demonstrate values entirely within reference ranges, indicating preserved hepatic synthetic and excretory function.

ML Result: Normal— the predictive model assesses the profile as within normal limits.

Detailed Test Explanations

Serum Bilirubin (Total)

: Measures total bilirubin, a byproduct of red blood cell breakdown processed by the liver; elevated levels suggest hemolysis, hepatic dysfunction, or biliary obstruction. Patient result: 0.9 mg/dl (reference: 0.2-1.2 mg/dl).

Serum Bilirubin (Direct)

: Quantifies conjugated bilirubin excreted in bile; increases indicate hepatocellular or obstructive jaundice. Patient result: 0.2 mg/dl (reference: 0-0.3 mg/dl).

Serum Bilirubin (Indirect)

: Reflects unconjugated bilirubin; elevations point to pre-hepatic causes like hemolysis or impaired conjugation. Patient result: 0.70 mg/dl (reference: 0.2-1.0 mg/dl).

SGPT (ALT)

: Alanine aminotransferase, a sensitive enzyme marker of hepatocellular injury; rises with hepatitis or toxin exposure. Patient result: 36 U/L (reference: 13-40 U/L).

SGOT (AST)

: Aspartate aminotransferase, found in liver, heart, and muscle; elevations signal acute liver injury or extrahepatic sources. Patient result: 32 U/L (reference: 0-37 U/L).

Serum Alkaline Phosphatase

: Enzyme elevated in biliary obstruction, bone disease, or pregnancy; no value available for assessment.

Serum Protein

: Total protein reflecting nutritional status and hepatic synthesis; low levels indicate malnutrition or chronic liver disease. Patient result: 7.2 g/dl (reference: 6.4-8.3 g/dl).

Serum Albumin

: Primary protein synthesized by the liver, indicating chronic liver function and nutritional adequacy; hypoalbuminemia suggests cirrhosis or nephrotic syndrome. Patient result: 4.7 g/dl (reference: 3.5-5.2 g/dl).

Globulin

: Immunoglobulin fraction; abnormalities reflect inflammation, infection, or dysproteinemia. Patient result: 2.50 g/dl (reference: 1.8-3.6 g/dl).

A/G Ratio

: Albumin-to-globulin ratio; derangements indicate chronic inflammation or liver synthetic defects. Patient result: 1.88 (reference: 1.1-2.4).

Overall Interpretation

All assessed liver function parameters fall within normal reference intervals, supporting intact hepatocellular integrity, bilirubin metabolism, and protein synthesis. No evidence of hepatitis, cirrhosis, or fatty liver disease. The

ML Result: Normal

corroborates this unremarkable profile, with clinical correlation advised per standard practice.

Recommendations

Maintain a balanced diet rich in fruits, vegetables, and lean proteins; limit alcohol intake and avoid hepatotoxins. Engage in regular aerobic exercise (150 minutes weekly) and monitor weight to prevent fatty liver. Routine annual liver function screening recommended; consult physician for symptoms like jaundice or abdominal pain.

Disclaimer

This report is for informational purposes only and does not constitute medical advice. Interpretations should be confirmed by a qualified healthcare provider in conjunction with full clinical evaluation.